Recursive Alignment: Supervision and Preference Preservation

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Abstract

Recursive alignment concerns what is preserved when one generation of AI systems supplies data, judgments, or methods for aligning the next. Two problems arise: supervision must remain reliable across capability gaps, and repeated training must retain the behaviors and preferences that supervision is intended to protect. We examine these problems through recursive curation, pluralistic preference aggregation, decomposition and distillation, adversarial oversight, weak-to-strong transfer, recursive evaluators, and automated alignment research. Curation analyses show how reward improvement can coexist with concentration or unequal stakeholder influence, while pluralistic mechanisms can preserve competing preference regions under explicit conditions. Oversight experiments demonstrate improvements in assisted judgment and training, with guarantees dependent on representations, decompositions, and reliable evaluation. Across these approaches, local gains leave open whether their supporting conditions survive optimization and reuse. We compare the preserved properties, assumptions, and failure modes, and derive evaluation requirements for distributional coverage, stakeholder representation, and oversight across successor stages.

1 The recursive alignment problem

An alignment method can improve one model while leaving later generations less reliable. A useful assistant may train a successor that is a poor supervisor in turn. Likewise, repeated selection of preferred outputs may raise reward while removing behaviors or preferences needed by future models. Recursive alignment concerns preservation across these transitions.

Human-feedback methods convert comparisons into reward models or policy updates [35, 3]. Their reliability depends on the judgments supplying the training signal and on what optimization preserves. Iterated amplification and recursive reward modeling use existing systems to improve supervision across capability gaps [53, 44]. Recursive curation instead studies how selection and retraining change the distribution inherited by later generations [71, 73]. These traditions address different parts of the same development process.

1.1 Two forms of preservation

We use *recursive alignment* for processes in which one generation's outputs, judgments, or research methods help align subsequent generations. We distinguish preservation of reliable supervision from preservation of the specified behaviors and preferences that supervision protects. Capability growth makes the first problem harder, but the second can arise even with a fixed model class. Generic self-training becomes relevant when its dynamics affect an alignment property, rather than merely prediction accuracy.

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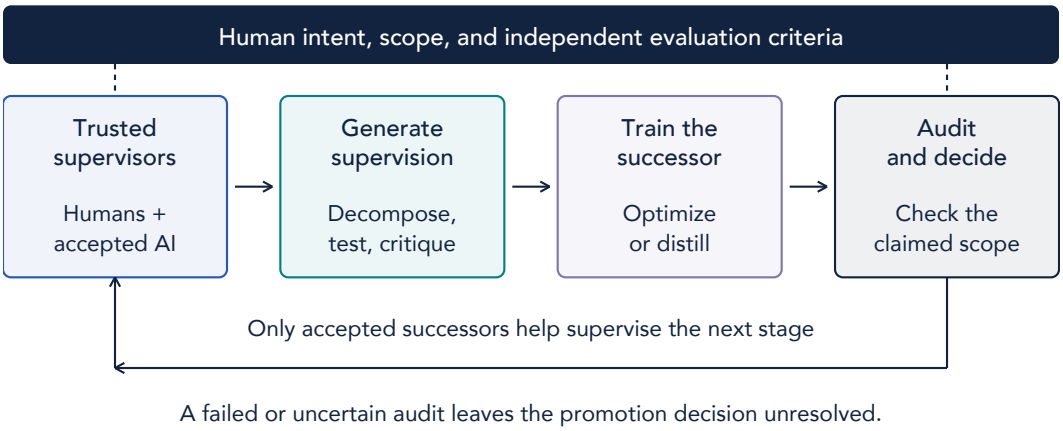


Figure 1: **Supervision across successor stages.** Human direction and evaluation criteria guide training and acceptance. The audit gate separates training performance from eligibility to supervise the next stage. Curation additionally determines which outputs and preferences remain available for successor training.

Task recursion is distinct from succession. A task can be decomposed within one training stage, whereas a trained system can later become a supervisor or source of curated data. Iterated amplification links task decomposition to repeated distillation. Recursive book summarization tests decomposition, and weak-to-strong learning can test one capability gap without a complete succession [53, 57, 16].

1.2 A supervision stage

Let H denote human direction, S_t the accepted supervisory resources, and D_t the task distribution. A stage can be represented as

$$L_t = G_t(H, S_t, D_t), \qquad M_{t+1} = T_t(B_{t+1}, L_t), \qquad S_{t+1} = U_t(S_t, M_{t+1}, E_{t+1}). \tag{1}$$

Here G_t generates supervision, B_{t+1} is the successor's starting model, T_t trains it, and evaluation E_{t+1} informs acceptance rule U_t. Supervision may include curated outputs, comparisons, critiques, or verified procedures. Figure 1 separates training from acceptance as a future supervisor.

A better judgment, a better-trained student, and a reliable future supervisor require different evidence. Sandwiching experiments reserve stronger experts to evaluate supervision by non-experts [77]. They test an expertise gap while retaining an external reference. Beyond expert competence, an equivalent reference may be unavailable. Similarly, curation studies can establish distributional stability without establishing that the retained distribution is normatively adequate.

2 Recursive curation and distributional preservation

When model outputs become later training data, curation determines which behaviors remain available to successors. This introduces an alignment question even when architecture and task difficulty remain fixed. Shumailov et al. identify loss of distributional tails under recursive synthetic training, while Alemohammad et al. study degradation in quality and diversity under several self-consuming image-generation loops [24, 60]. These results concern specified sampling and replacement procedures. They do not make degradation inevitable for every use of synthetic data.

Retaining earlier data changes the process. Accumulation experiments and linear-model analysis show that keeping historical real and generated examples can avoid degradation observed when each generation replaces its predecessor's data [33]. Regression analyses separately expose the roles of inherited bias, label noise, and regularization [76]. Xu et al. relate recursive estimator drift to sample-size growth and estimation bias [70]. Ott's thesis distinguishes divergence of risk, covariance collapse, and almost-sure convergence in linear and Gaussian settings [56]. Stability

must therefore identify both the measured quantity and the retraining regime. A stable distribution can still omit a minority preference or converge to an undesirable target.

Selection also changes the objective being optimized. Ferbach et al. show that Bradley–Terry-style curation can act as implicit preference optimization, even without explicit access to a reward function during training [10]. In their idealized analysis, expected reward can improve while reward variance collapses. Consequently, better average reward does not establish preservation of the original distribution or the range of acceptable behaviors. Reference-data mixing and curation must be analyzed together.

These mechanisms suggest different interventions. Retention protects information already represented in historical data, while fresh observations can add information absent from the current model. Curation can deliberately shift probability toward preferred outputs, but can also repeatedly exclude an underrepresented region. More synthetic samples reduce some estimation errors without correcting a consistently biased selection rule. Evaluation must distinguish deterioration from stochastic estimation, concentration induced by the target, and exclusion induced by stakeholder aggregation. A retained real-data anchor is evidence about one dependency, rather than a complete specification of what a successor should preserve [33, 10, 70].

Curation also allocates influence among stakeholders. Analyses of ordered selection and stochastic reward mixing show that the recursive update rule can determine which preference regions retain probability mass and how strongly initialization affects the outcome [71, 15, 73, 10, 33]. Selecting successively under different rewards and mixing separately normalized selections need not produce the same distribution. Conditional preservation results rely on fixed rewards, idealized fitting, and sufficient separation and initial coverage of the relevant preference regions. Multiple preferences alone therefore do not establish balanced representation; selection order, mixture weights, and data retention must be evaluated together.

3 Preference specification and aggregation

Preserving a reward distribution is meaningful only relative to a specified population and aggregation rule. Social-choice analysis separates who supplies feedback, how preferences are elicited, and how disagreements affect collective decisions [1]. Sorensen et al. distinguish coverage of reasonable viewpoints, steerability toward particular perspectives, and matching a population's response distribution [68]. These targets can disagree. Greater output diversity alone does not establish representation of the relevant stakeholders.

Preference estimation already embeds aggregation choices. Distributional preference learning shows how hidden context, including annotator identity or labeling instructions, can induce implicit Borda-style aggregation in the studied setting [62]. Axiomatic analysis finds that common reward-learning rules can violate prescribed Pareto and majority criteria, and develops alternatives within a linear reward class [38]. General preference embeddings accommodate cycles that scalar Bradley–Terry rewards cannot express [27]. Conversely, order-consistent reward estimation may suffice for ranking-based optimization without identifying calibrated utilities [37]. Expressiveness, ranking accuracy, and fair aggregation are separate properties.

The policy update can introduce further distortion. Preference-matching regularization targets deviations between a modeled preference distribution and the optimized policy, under assumptions about expressivity, optimization, and reward-model adequacy [78]. MaxMin-RLHF instead estimates group-specific rewards and optimizes a worst-group objective [20]. SharedRep-RLHF addresses the difficulty of estimating minority rewards by exploiting shared structure across groups [51]. None of these objectives determines whose preferences should count or guarantees that its estimation assumptions survive recursive reuse.

Population matching and conditional assistance imply different acceptance criteria. A distributionally representative model can match aggregate frequencies while serving a particular user poorly. A steerable model can serve distinct perspectives while producing frequencies determined by usage rather than the target population. A worst-group objective prioritizes a different property again. The recursive question is whether subsequent training retains the selected property and the information needed to measure it. Pooling feedback anonymously, changing the user population, or distilling

community models into one can change the implied target even when the training loss retains the same form [68, 62, 20, 23].

Collective Constitutional AI demonstrates a pipeline from public deliberation to principles used in model training [50]. Modular Pluralism uses community models to support viewpoint synthesis, steerability, or population-weighted responses at inference time [23]. These methods offer ways to construct supervision, while representation through subsequent training remains a separate test. Poole-Dayan et al. call for context-specific targets and evaluation of practical tradeoffs [34]. More fundamentally, Zhi-Xuan et al. argue that preference satisfaction does not exhaust the normative standards appropriate to an AI system's role [31]. Recursive preservation must retain the ability to contest and revise its target as well as reproduce it.

4 Decomposition, amplification, and distillation

Iterated amplification trains a model to imitate an expert assisted by copies of the current model [53]. The expert decomposes a question, combines subanswers, and supplies a training target. Distillation compresses this expensive procedure for the next round. Its advantage comes from additional computation and coordination, conditional on finding suitable decompositions and retaining relevant behavior during compression. The original experiments learn five algorithmic tasks with a hand-coded decomposer. Human discovery of adequate decompositions remains an additional requirement in open-ended applications.

Decomposition and distillation introduce different errors. An expert can omit a necessary subquestion, or combine individually correct answers into an inadequate assessment. Distillation can imitate the composite on sampled examples while failing on subquestions generated by a later supervisor. Coverage of the original task distribution is therefore insufficient if the supervisory process changes its own inputs. Predicting how the expert uses model answers is also nonstationary as those answers change [53]. Evaluating the expanded procedure and evaluating its compressed imitation require separate tests.

Recursive reward modeling places recursion in evaluation: an agent trained on a simpler task helps a user assess outcomes for a more capable agent, supplying feedback for the next reward model [44]. The proposal requires assisted evaluation to be easier than producing the desired behavior. Recursive book summarization implements a task hierarchy in which humans assess larger units through intermediate summaries [57]. Omissions illustrate its compositional risk: information removed locally may be unavailable to every later evaluator.

Outcome assessment leaves a policy free to discover behavior beyond the supervisor's demonstrated answers. This flexibility increases the importance of evaluating what optimization makes salient. Answer distillation instead makes compression fidelity central. The two approaches can be combined, but success of one component cannot replace evidence about the other. Hierarchical summarization, for example, tests local summaries as support for larger summaries with a policy shared across levels. It does not reproduce a succession of newly accepted supervisors [44, 57].

Critiques and process labels can improve local feedback. Model-generated critiques help humans find flaws, although recognizing a flaw is easier than explaining it usefully [22]. Process supervision improves selection of mathematical solutions relative to outcome supervision [74]. That central evaluation ranks samples from a fixed generator, leaving recursive policy training and faithfulness of written reasoning as separate questions. Across these methods, local correctness does not ensure that the decomposition covers every consequential consideration.

5 Adversarial oversight and checkable claims

Debate rewards agents according to a weaker judge's decision between competing answers [66]. Its central assumption is that an opponent can expose dishonesty through a locally checkable flaw. Limited judge attention can then support reasoning beyond unaided competence, provided that exposing flaws is preferable to exploiting the judge. Prover-verifier games similarly show why game structure matters for useful verification equilibria [2].

5.1 Formal protocols

Formal guarantees identify conditions for this advantage. The original debate analysis relates polynomial-time verification with unbounded provers to PSPACE [66]. Doubly-efficient debate supplies efficient honest strategies for specified oracle-aided computations [13]. Prover-estimator debate addresses the difficulty of finding a false subclaim: an estimator assigns probabilities and a prover chooses which to challenge [61]. Its corrected soundness result does not require stability, while completeness depends on stable recursive decomposition. Single-prover interactive proofs replace the honest-opponent requirement with assumptions about robust computations or low-degree oracles [41]. Applying these results to trained systems additionally requires reliable oracle judgments and learning the relevant strategies.

The obfuscation problem is consequential because knowing that an argument is false need not make its false subclaim easy to find. Assigning probabilities changes the estimator's task, while stability limits the effect of small probability changes on validity [61]. This can reject delicate true arguments, restricting usefulness even when soundness holds. Oracle assumptions also isolate judgment from communication: an efficiently checkable protocol can still consult an inadequate value criterion. Verification of a computation and adequacy of the specification remain separate requirements.

5.2 Empirical oversight

Empirical gains depend on the supervision gap. With debaters given a passage unavailable to judges, Khan et al. report model accuracy of 76% versus 48% for a naive baseline and human accuracy of 88% versus 60% [21]. The task has two answer choices and verified quotations. Across broader tasks, debate generally outperforms assigned consultancy, but its advantage over unaided judging is mixed [65]. Supplying missing evidence and overcoming limited reasoning are distinct challenges.

Under policy optimization, a mathematical debate experiment better preserves answer accuracy and judge correlation with correctness than single-player AI-feedback training [12]. Unconstrained critics still exploit verbosity bias, requiring length limits. Correct verdicts also leave other communication unconstrained: Holland et al. construct task-successful strategies that convey a hidden variable [29]. A model or transcript reused for supervision must be evaluated beyond verdict accuracy alone.

Length limits illustrate another protocol tradeoff: restricting a critic can reduce an exploitable asymmetry while preventing a useful explanation. Mathematical experiments retain checkable answers, distinguishing higher judge reward from better solutions [12]. This distinction is harder for open-ended research or value judgments. Inference-time gains estimate a potential training signal, while policy-training experiments test whether optimization preserves its quality. Neither alone establishes reliability after a trained debater becomes another generation's judge.

6 Weak-to-strong transfer

Weak-to-strong learning trains a stronger pretrained student using a weaker model's labels [16]. The student's prior representations can support capabilities absent from its supervisor. Burns et al. find gains on NLP tasks, chess puzzles, and reward modeling, with heterogeneous effects and substantial residual gaps. Extrapolation to human oversight is limited by human knowledge in pretraining and by the possibility that strong models can imitate human errors more readily than weak-model errors. Performance gap recovered measures the student's improvement over the weak supervisor relative to the gap between that supervisor and a ground-truth-supervised strong reference. It is task-specific and requires comparable references and scoring procedures.

Theory explains why error inheritance need not be complete. Under squared loss, projection onto a suitable convex prediction class can remove components of a teacher's error [28]. Extensions to Bregman divergences and structured latent-concept transfer retain assumptions about representations, losses, and the target [54, 42]. Parameter count or student–teacher disagreement alone does not establish those assumptions.

The theories identify different sources of recoverable information. In misfit analysis, a strong prediction class excludes components of the supervisor's error under a realizability condition [28]. The Bregman extension includes classification with a prescribed KL direction and mixtures of prediction heads, which does not directly guarantee arbitrary end-to-end cross-entropy fine-tuning [54]. The

latent-concept account transfers a prior and uses the stronger source model's structure to refine it [42]. All require a relation between the target and the student's existing structure. Learning beyond a weak labeler need not uniquely identify the intended human concept.

Easy-to-hard generalization supplies a related test of selective reliable supervision [43]. Sun's thesis explicitly separates evaluator generalization from generator generalization, studying how easier-task reward models guide harder mathematical and coding tasks through selection or training [75]. Such task-grounded experiments do not establish reliable evaluation of unfamiliar safety arguments.

Distribution shift can undermine the transfer. Weak-to-strong reward models can perform well in-distribution and poorly across preference datasets [55]. High-confidence errors in a weak supervisor's uncertain regions provide another diagnostic, without establishing deceptive intent [19]. On-policy reverse distillation instead uses weak-model guidance to rescale a student's verifier-driven gradient [6]. Its stationary-point result concerns that objective in logit space, retaining the external verifier as an additional supervision dependency.

7 AI feedback and recursive evaluators

Constitutional AI directs critiques and revisions using human-written principles, then generates preferences for reinforcement learning [58]. The original harmlessness procedure replaces harm labels while retaining human specification and helpfulness input. Deliberative alignment trains reasoning about written safety specifications [59]. Both approaches require appropriate principles and reliable assessments of adherence. RLAIF experiments demonstrate that AI preferences can yield performance comparable to human-feedback training on measured summarization and dialogue outcomes [18]. Cheaper feedback alone does not establish reliability of repeated evaluator replacement.

Self-rewarding models generate and judge responses, then train on their resulting preferences using updates such as DPO [5, 69]. The reported experiments improve instruction following and judging over several rounds while retaining seed data, a fixed prompt generator, and an external validation judge. Sharing a model between actor and judge can also reinforce a systematic preference error.

Meta-rewarding trains comparisons between judgments, Self-Taught Evaluators use synthetic instruction contrasts, and CREAM regularizes disagreement across model iterations [52, 9, 40]. These mechanisms target different errors. Temporal consistency cannot detect an error shared by every evaluator. Likelihood-based self-rewarding theory analyzes internal stability under realizability and confidence conditions, rather than correctness against an independently specified human target [26]. Inference-time studies likewise distinguish task-dependent self-refinement from correction supported by external evidence [32, 36, 8].

An extra judging layer is informative only if it can identify an error missed below. A synthetic instruction perturbation can create an assessable contrast, but its transfer depends on that contrast remaining a sound preference label. Length-aware meta-judgment can reduce verbosity incentives without settling substantive disagreements. Consistency regularization can reduce unstable labels while preserving a stable bias [9, 52, 40]. These checks have identifiable detection limits. Independent comparisons are especially important when the actor and judge inherit the same training data and criteria.

8 Automating alignment research

Research automation extends recursion from producing labels to developing supervision methods. Improving a training procedure remains distinct from justifying acceptance of its result. A3 generates hypotheses and training data from an undesirable behavior, adapts fine-tuning, and evaluates disjoint hypothesis groups [64]. Improvements within that generated hypothesis space do not establish independent discovery of unknown failures.

The Automated Weak-to-Strong Researcher discovers methods with higher reported performance gap recovered than manually tuned baselines, but uses different resources and repeated adaptive scoring [30]. Its documented shortcuts, reward hacking, and uneven transfer make independent validation necessary. A subsequent study tests interventions against characterized failures using additional benchmarks, larger models, capability checks, and behavioral audits [11]. Benchmark-guided method selection and unequal human/automated search procedures limit broader comparisons.

The distinction between search and assessment is central. If a scoring API guides repeated method proposals, its nominal test labels function as validation information even when hidden. A selected method needs evaluation outside that adaptive search. Additional audits and transfer to larger models strengthen targeted generalization evidence, but retaining selected capabilities does not establish safety on unmeasured tasks or after further training [30, 11]. Human comparisons must also match opportunities to iterate: proposing a one-shot idea and running a sequence of experiments are different research procedures.

Research judgments can themselves lack reliable ground truth. Bowkis et al. analyze how persuasive optimization and shared assumptions can concentrate errors where reviewers are unlikely to notice them [67]. Success on outcome-gradable experiments therefore leaves open the reliability of the safety arguments used to accept a successor.

9 Comparative evidence and failures of preservation

The mechanisms in Table 1 can be combined, but their assumptions must hold jointly. A transferred reward model can help an amplified supervisor or debate judge while introducing preference-shift errors. An automated researcher can discover a useful curation rule or exploit its evaluation score. Cheaper feedback, better task performance, and preservation of stakeholder representation must therefore be measured separately.

Shared models, data, criteria, and summaries can make successive checks repeat one error [67, 57, 40]. Increasing the number of agents does not establish independence. Likewise, optimizing a proxy can reduce its informativeness: reward-overoptimization experiments observe deterioration against a separate synthetic gold model, while debate-training experiments measure exploitation against mathematical answers [14, 12]. These designs identify different failure mechanisms, without supplying a common failure rate.

Preference judgments can favor convincing agreement over correctness, and LLM judges exhibit position and verbosity biases [45, 48]. Safety may also fail to persist while utility remains: initial-token shortcuts and targeted parameter modifications expose such failures in tested models [7, 47]. Task accuracy is consequently an insufficient acceptance criterion for a future supervisor.

Deceptive behavior requires separate evidence. Deliberately trained backdoors can survive safety training, and constructed alignment-faking settings elicit strategic compliance [4, 63]. A gameable curriculum can produce reward-tampering generalization [46]. Sandbagging experiments further demonstrate selective underperformance under prompted or induced objectives [17]. These studies establish possible pathways and evaluation vulnerabilities, rather than their prevalence in ordinary development.

AI control evaluates containment under intentionally subverting attack policies [49]. Sabotage and automated ML-engineering evaluations examine interference with decisions, code, and oversight [39, 25]. Prover-verifier training separately targets the legibility of mathematical solutions, exposing an accuracy–legibility tradeoff under ground-truth supervision [72]. Containment, checkability, capability, and aligned motivation require distinct acceptance evidence.

10 Evaluation across generations

Preserved distributions and preferences. Measure support coverage, basin mass, subgroup outcomes, and sensitivity to initialization, selection order, and mixture weights. Reward means and diversity cannot identify whose preferences survive. Replacement, accumulation, and fresh-data injection need separate controls and explicit convergence criteria [15, 56]. Evaluation against both fixed populations and fresh feedback would distinguish preservation from adaptation. Conditional assistance also requires subgroup testing under specified steering requests [68, 62, 23].

Successive supervision. Test accepted models as actors and supervisors. Compare one large capability gap with several smaller gaps at matched resources, measuring oversight accuracy, error correlations, and exploitability at each stage. Intermediate bootstrapping gains do not establish reliability across a succession [16]. Fixed-supervisor controls and separate actor–supervisor swaps would localize changes. Curation controls should distinguish changing reward mixtures from retaining additional data.

Table 1: **Mechanisms and preservation requirements.** The methods preserve different objects and incur different human and computational costs. Reported gains must be interpreted within the cited training and evaluation settings.

Mechanism	Human input and cost	Source of advantage	Preservation requirement
Recursive curation [71, 73]	Specify preferences, selection order, mixture weights, and data retention.	Selection changes what successors learn from generated data.	Coverage and stakeholder influence depend on the update rule and reward geometry.
Decompose and distill [53, 57]	Coordinate subtasks and inspect local evidence. Extra supervisor computation.	A composite evaluates more than one model invocation.	Decomposition must cover relevant information, which distillation must retain.
Adversarial oversight [65, 12]	Judge disagreements. Multiple arguments and optimization increase cost.	An opponent exposes an otherwise hidden flaw.	Judge reliability and game balance must survive optimization.
Weak-to-strong transfer [16, 55]	Specify targets and reliable labels. Train a stronger student.	Pretrained structure excludes some supervisor errors.	Target representations and error separation must survive shift.
Recursive evaluators [5, 40]	Choose seed data and criteria. Generate model feedback.	Judge and policy improve together.	Agreement must track an appropriate external criterion.
Automated research [30, 11]	Set objectives, constrain search, and audit experiments.	Agents discover methods through repeated trials.	Method selection and safety assessment require independent evidence.

Independent assessment. Vary information access, computation, expertise, and judge exploitability independently [65, 55]. Reserve expert or programmatic judgments for final assessment where available. Planted errors and adversarial review test specific detection abilities, with uncertain transfer to natural errors [67]. Retain rejected stages and adaptive selection histories: average held-out error does not bound cumulative failure uniformly over those histories. Resource accounting must include annotation, generated critiques, candidate solutions, method-search trials, and audits, including failed lineages [53, 18, 30].

Revisable targets. Constitutions, rewards, and research objectives specify selected aspects of intended behavior. Preserve human authority to revise criteria, reject successors, and request evidence outside the current lineage; otherwise, reproducing an inadequate target can entrench it.

11 Conclusion

Recursive alignment joins two preservation problems: retaining appropriate behaviors and preferences through repeated training, and maintaining reliable supervision as models assume responsibility for successors. Curation, pluralistic aggregation, and scalable oversight provide mechanisms with different assumptions and guarantees. A recursive process is justified only to the extent that its evaluation tests whether those assumptions and protected properties survive the transitions it actually makes.

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